Residue Theorem

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

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Abstract

A complete authored walkthrough of residue theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

CA-03 / Complex Analysis / Report v1

I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let D be a bounded Jordan domain whose positively oriented boundary gamma is piecewise smooth. Let f be holomorphic in an open neighborhood of the closure of D except for finitely many isolated singularities $a_1, \ldots, a_s$ inside D. Then*

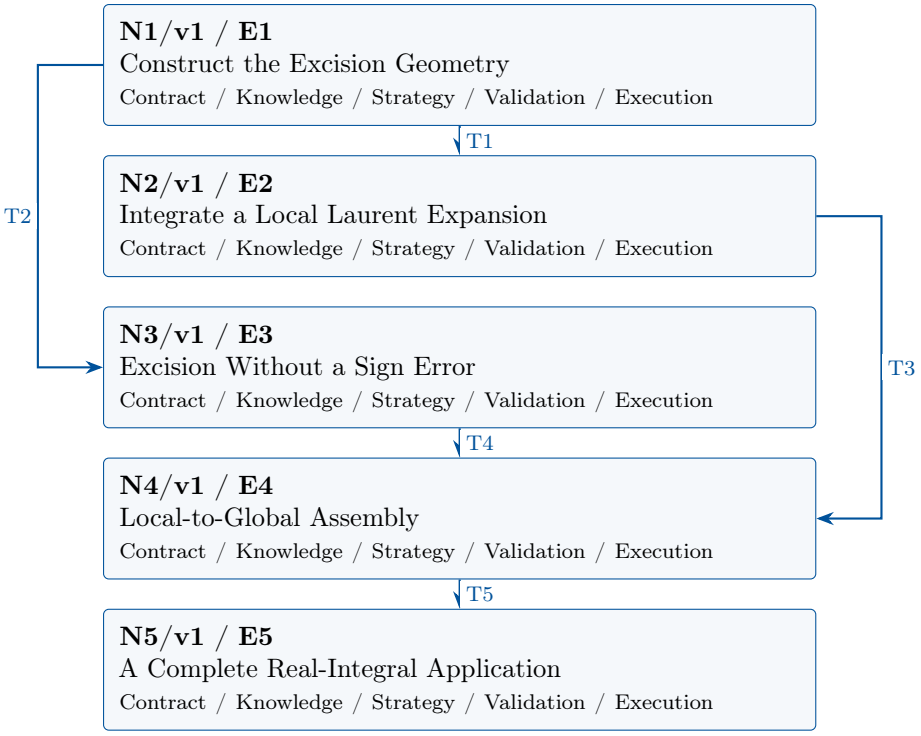
$$\oint_\gamma f(z)\,dz = 2\pi i \sum_{j=1}^{s} \operatorname{Res}(f, a_j).$$

The output is a contour identity with all interior singularities included. Singularities on the contour are excluded. Essential isolated singularities are allowed; the theorem does not require every singularity to be a pole.

A. Foundations and permitted imports

a residue is the coefficient of $(z-a)^{-1}$ in the Laurent expansion; positive outer orientation is counterclockwise. Import the Laurent expansion theorem and uniform convergence on compact annuli. Import Cauchy–Goursat's boundary form: a function holomorphic near a finitely connected region's closure has zero integral around its full oriented boundary. This follows by cuts or subdivision into simply connected pieces with cancellation of internal sides; it is not the residue theorem being proved.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. Every isolated interior singularity has a disjoint closed isolating disk lying in the domain.

Receiving activity. N2: Integrate a Local Laurent Expansion

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and the Laurent expansion on each punctured disk, uniformly convergent on the circle.

T2: N1/E1/v1 to N3. Every isolated interior singularity has a disjoint closed isolating disk lying in the domain.

Receiving activity. N3: Excision Without a Sign Error

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and Cauchy–Goursat for the region with the disks removed.

T3: N2/E2/v1 to N4. $\oint_{C_j} f(z)\,dz = 2\pi i \operatorname{Res}(f, a_j)$ for each counterclockwise local circle.

Receiving activity. N4: Local-to-Global Assembly

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2 for every singularity and N3/E3 for the same contour.

T4: N3/E3/v1 to N4. $\oint_\gamma f = \sum_j \oint_{C_j} f$.

Receiving activity. N4: Local-to-Global Assembly

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2 for every singularity and N3/E3 for the same contour.

T5: N4/E4/v1 to N5. $\oint_\gamma f(z)\,dz = 2\pi i \sum_j \operatorname{Res}(f, a_j)$.

Receiving activity. N5: A Complete Real-Integral Application

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N4/E4 for upper semicircles of radius $R > 1$; elementary improper integration.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Construct the Excision Geometry

Object and role. Construct the Excision Geometry. Establish the following local result: Every isolated interior singularity has a disjoint closed isolating disk lying in the domain.

Inputs. A positively oriented piecewise smooth Jordan boundary; finitely many isolated interior singularities and none on the boundary.

C — Contract

Own E1/v1: Every isolated interior singularity has a disjoint closed isolating disk lying in the domain.

K — Knowledge

Holomorphic domain and singularities

S — Strategy

Choose disjoint local disks

V — Validation

Exclude boundary singularities

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Use finiteness and interiority to choose disjoint small disks.

N1.2 Keep their boundaries away from singularities.

N1.3 Record the outer orientation and the counterclockwise local-circle conventions.

Export E1/v1. Every isolated interior singularity has a disjoint closed isolating disk lying in the domain.

Acceptance gate. Exclude boundary singularities Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Exclude boundary singularities Recheck dependent nodes N2, N3, N4, N5.

B. N2/v1: Integrate a Local Laurent Expansion

Object and role. Integrate a Local Laurent Expansion. Establish the following local result: $\oint_{C_j} f(z)\,dz = 2\pi i \operatorname{Res}(f, a_j)$ for each counterclockwise local circle.

Inputs. N1/E1 and the Laurent expansion on each punctured disk, uniformly convergent on the circle.

C — Contract

Own E2/v1: $\oint_{C_j} f(z)\,dz = 2\pi i \operatorname{Res}(f, a_j)$ for each counterclockwise local circle.

K — Knowledge

N1/E1 and the Laurent expansion on each punctured disk, uniformly convergent on the circle.

S — Strategy

Integrate a Local Laurent Expansion.

V — Validation

Check the hypotheses and each step of the displayed calculation.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Integrate the Laurent series term by term on the compact circle.

N2.2 Parametrize $z = a_j + re^{it}$ and integrate each integer power.

N2.3 Only the coefficient of $(z - a_j)^{-1}$ contributes; poles and essential isolated singularities are both covered.

Export E2/v1. $\oint_{C_j} f(z)\,dz = 2\pi i \operatorname{Res}(f, a_j)$ for each counterclockwise local circle.

Acceptance gate. Check the hypotheses and each step of the displayed calculation. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Check the hypotheses and each step of the displayed calculation. Recheck dependent nodes N4, N5.

C. N3/v1: Excision Without a Sign Error

Object and role. Excision Without a Sign Error. Establish the following local result: $\oint_\gamma f = \sum_j \oint_{C_j} f$.

Inputs. N1/E1 and Cauchy–Goursat for the region with the disks removed.

C **Contract**

Own E3/v1: $\oint_\gamma f = \sum_j \oint_{C_j} f$.

K **Knowledge**

Cauchy–Goursat boundary form

S **Strategy**

Remove singularity disks

V **Validation**

Track clockwise hole boundaries

E **Execution**

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Apply the boundary theorem where f is holomorphic.

N3.2 Inner boundaries are clockwise in the punctured domain.

N3.3 Move their negatives to the other side, leaving counterclockwise local integrals.

Export E3/v1. $\oint_\gamma f = \sum_j \oint_{C_j} f$.

Acceptance gate. Track clockwise hole boundaries Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Track clockwise hole boundaries Recheck dependent nodes N4, N5.

D. N4/v1: Local-to-Global Assembly

Object and role. Local-to-Global Assembly. Establish the following local result: $\oint_\gamma f(z)\,dz = 2\pi i \sum_j \operatorname{Res}(f, a_j)$.

Inputs. N2/E2 for every singularity and N3/E3 for the same contour.

C — Contract

Own E4/v1: $\oint_\gamma f(z)\,dz = 2\pi i \sum_j \operatorname{Res}(f, a_j)$.

K — Knowledge

All local certificates

S — Strategy

Sum independent local contributions

V — Validation

Include every singularity

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Substitute every local residue integral into the excision identity.

N4.2 Check no singularity is omitted or counted twice.

N4.3 With no singularities the empty sum recovers Cauchy–Goursat.

Export E4/v1. $\oint_\gamma f(z)\,dz = 2\pi i \sum_j \operatorname{Res}(f, a_j)$.

Acceptance gate. Include every singularity Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Include every singularity Recheck dependent nodes N5.

E. N5/v1: A Complete Real-Integral Application

Object and role. A Complete Real-Integral Application. Establish the following local result: $\int_{-\infty}^{\infty}(1+x^2)^{-1}\,dx = \pi$.

Inputs. N4/E4 for upper semicircles of radius $R > 1$; elementary improper integration.

C — Contract

Own E5/v1: $\int_{-\infty}^{\infty}(1+x^2)^{-1}\,dx = \pi$.

K — Knowledge

Contour identity and decay estimates

S — Strategy

Separate real and arc integrals

V — Validation

Audit improper limit

E — Execution

Carry out the numbered deductions and publish E5/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N5.1 The sole enclosed pole is i, with residue $1/(2i)$.

N5.2 Bound the arc integral by $\pi R/(R^2-1)$, which tends to zero.

N5.3 Use absolute integrability on the real axis and take the radius limit.

Export E5/v1. $\int_{-\infty}^{\infty}(1+x^2)^{-1}\,dx = \pi$.

Acceptance gate. Audit improper limit Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N5: Audit improper limit Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Construct the Excision Geometry

Choose small closed disks around the a_j, disjoint from one another and contained in D, with no other singularity in any disk. Let C_j be the positively oriented circle bounding the jth disk. Their radii can be chosen because the singularity list is finite and each point is isolated.

B. N2 — Integrate a Local Laurent Expansion

Write $f(z) = \sum_{k\in\mathbb{Z}} c_k(z-a_j)^k$ in the punctured local neighborhood. Uniform convergence on C_j justifies termwise integration. Parameterizing $z = a_j + re^{it}$ yields

$$\oint_{C_j} (z-a_j)^k\,dz = \begin{cases} 2\pi i & k=-1, \\ 0 & k\neq -1. \end{cases}$$

For $k \neq -1$ the exponential $e^{i(k+1)t}$ integrates to zero; for $k = -1$ the integrand times dz is i dt. Therefore $\oint_{C_j} f = 2\pi i c_{-1} = 2\pi i \operatorname{Res}(f, a_j)$.

C. N3 — Excision Without a Sign Error

Remove the disk interiors from D. The resulting region has boundary gamma together with the hole circles in clockwise orientation. Cauchy–Goursat applies because every singularity has been excised. Hence

$$0 = \oint_\gamma f - \sum_j \oint_{C_j} f.$$

Thus the outer contour integral equals the sum of the counterclockwise local integrals. An inner circle's orientation is reversed in the region boundary; the minus sign is part of the proof.

D. N4 — Local-to-Global Assembly

Substitute the N2 certificates into the N3 identity. This gives exactly the theorem. With no singularities, the sum is empty and the result reduces to Cauchy–Goursat. □

E. N5 — A Complete Real-Integral Application

For $f(z) = 1/(1+z^2)$, take the real segment from $-R$ to R followed by the upper semicircle, $R > 1$. The only interior pole is i, with

$$\mathrm{Res}(f, i) = \lim_{z \to i} \frac{z - i}{(z - i)(z + i)} = \frac{1}{2i}.$$

The contour integral is therefore pi. On the arc, $|1 + z^2| \geq R^2 - 1$ and its length is pi R, giving

$$\left| \int_{\mathrm{arc}} \frac{dz}{1 + z^2} \right| \leq \frac{\pi R}{R^2 - 1} \longrightarrow 0.$$

The real integrand is absolutely integrable: it is bounded near zero and at most $1/x^2$ for $|x| \geq 1$. Passing to the improper limit gives

$$\int_{-\infty}^{\infty} \frac{dx}{1 + x^2} = \pi.$$

A residue calculation alone would not justify this conclusion without the arc estimate and existence of the real improper integral.

□

V. WORKED ORCHESTRATION

Take $f(z) = (z + 1)/((z - 1)(z - 2))$ on $|z| = 3$, positively oriented. The two local agents independently obtain

$$\mathrm{Res}(f, 1) = \frac{2}{-1} = -2, \qquad \mathrm{Res}(f, 2) = \frac{3}{1} = 3.$$

N4 sums them to 1, so the contour integral is $2\pi i$. Both poles must be included even though their contributions partially cancel.

- **Interaction**

  N1 → local agents

  Evidence transferred

  Disjoint analytic neighborhoods at 1 and 2

- **Interaction**

  Local agent 1 → N4

  Evidence transferred

  Residue −2 with simple-pole calculation

- **Interaction**

  Local agent 2 → N4

  Evidence transferred

  Residue 3 with simple-pole calculation

- **Interaction**

  N3 → N4

  Evidence transferred

  Outer integral equals sum of positively oriented local integrals

- **Interaction**

  N4 → caller

  Evidence transferred

  Total residue 1 and contour integral $2\pi i$

A. Boundary and hypothesis audit

- Reversing the outer orientation reverses the integral's sign.
- Higher-order poles still use the Laurent coefficient, not the simple-pole limit formula.
- A removable singularity has residue zero.
- Arc decay must be proved for the chosen function and contour.
- A plot or sampled singularity search does not establish the exhaustive pole list.

Export the local Laurent-integral lemma, the excision identity, the finite-singularity residue theorem, and the rational-integral application. These modules support the argument principle and Rouché's theorem in the problem bank.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

Suppose a branch uses $|z| = 1$ for $1/(1 + z^2)$. Both poles lie on the contour; the theorem's contract fails.

- **Node:** N1
- **State:** STALLED
- **Failing component:** Validation
- **Failed hypothesis:** no singularities on contour
- **Observed:** poles i and -i on $|z| = 1$
- **Repair target:** contour selection

- **Blocked export:** ordinary residue-theorem contour integral

Knowledge retrieves the domain hypothesis. Validation identifies both boundary poles. Strategy selects $|z| = 2$ for a revised problem. Execution checks both poles are interior and computes residues $1/(2\mathrm{i})$ and $-1/(2i)$, whose sum is zero. Contract exports integral zero for the revised contour. An indented contour or principal value would be a different problem requiring additional analysis.

B. Hypothesis boundary

- Reversing the outer orientation reverses the integral's sign.
- Higher-order poles still use the Laurent coefficient, not the simple-pole limit formula.
- A removable singularity has residue zero.
- Arc decay must be proved for the chosen function and contour.
- A plot or sampled singularity search does not establish the exhaustive pole list.

Export the local Laurent-integral lemma, the excision identity, the finite-singularity residue theorem, and the rational-integral application. These modules support the argument principle and Rouché's theorem in the problem bank.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Harvard complex analysis, Laurent series and residue theorem lectures.